

Nhan-An Tran, PhD

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PROFESSIONAL SUMMARY

Marine scientist and educator with expertise in microbiology, biotechnology and genomics, combined with a strong record of teaching, mentoring, and outreach. Experienced in developing educational resources, leading workshops, and supervising students in both international field and teaching experience across the US, UK, and Australia, and a proven ability to adapt to diverse cultural and educational contexts. Adaptable and collaborative team member with a passion for making science accessible through hands-on learning and community engagement.

EDUCATION

University of Cambridge , Ph.D. Plant Sciences	2023
University of Technology Sydney , B.Sc. (Honours) in Environmental Science	2017
University of Technology Sydney , B.Sc. Marine Biology	2016

CORE SKILLS

- ⇒ **Teaching:** Versatile teaching environments including small-group supervisions (3-5 students), laboratory teaching assistant (10-20 students) in Plant Biochemistry, Chemistry, Statistical Analysis and Design, Microbial Ecology, Chemical Oceanography
- ⇒ **Mentorship & Student Outreach:** Science mentoring and workshops for high-school students transitioning to university, community-orientated science communication
- ⇒ **Educational Resource Development:** Creation of teaching modules to supplement course materials, field and lab protocol development, computational training workshops
- ⇒ **Leadership & Collaboration:** Team management (Wolfson College May Ball President, 20-person team), cross-disciplinary project coordination at the college and council level
- ⇒ **Field & Laboratory Skills:** excellent laboratory skill set, with a focus on molecular biology and biotechnology, and at sea sampling capabilities

TEACHING, OUTREACH AND RESEARCH EXPERIENCE

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| Massachusetts Institute of Technology, Postdoctoral Associate | 10/2023 - Present |
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- Department of Earth, Atmospheric and Planetary Sciences (EAPS)
- ⇒ Co-facilitator for LEAPS: Leadership and Professional Strategies and Skills (Advancing your Professional Strategies and Skills and Developing your Leadership Competencies)
 - ⇒ Taught undergraduate and graduate students in Chemical Oceanography, providing laboratory guidance and fieldwork mentorship to develop key experimental skills
 - ⇒ Designed and delivered training materials for graduate students in bioinformatics and microbial genomics to develop computational biology skills
 - ⇒ Conducted at-sea outreach by engaging with students in oceanographic research activities aboard research vessels (SEA Education)

University of Cambridge, Ph.D

2019-2023

Department of Plant Sciences

- ⇒ Designed and delivered small-group supervisions (3-5 students) in Plant Biogeochemistry, providing personalized, discussion-based instruction and in-depth familiarity with course materials
- ⇒ Led laboratory demonstrations for undergraduate courses to help students gain confidence in wet lab skills, developed and adapted lab-based teaching materials as needed
- ⇒ Designed and delivered lectures and lab tours for visiting student groups, promoting microbial ecology to a broad audience

University of Technology Sydney, Teaching & Research Assistant

2013-2018

Department of Science, Climate Change Cluster

- ⇒ Taught laboratory classes (10-20 students) in Chemistry, Microbial Ecology and Statistical Analysis and Design, including developing assignments, grading and providing feedback
- ⇒ Science Outreach Team, designed and delivered student outreach activities for prospective students, school visits and off-campus activities
- ⇒ Mentoring of new students, helping them to build the laboratory skills, critical thinking and independence needed as scientific researchers

SELECTED PUBLICATIONS

Tran, N. A. T., Zhao, R., Cinay, T, Ward, BB., & Babbini, A.R. (manuscript in prep.). Size-driven redox microenvironments sustain anaerobic cycling in marine particles

Jacobi, Y., **Tran, N. A. T.**, Babbini, A.R., (manuscript in prep.). Light-Mediated Oxygen Dynamics and Denitrification in Marine Snow.

Tran, N. A. T., Mehrshahi, P., Archer, L., Faessler, C., Geisler, K & Smith, AG. (manuscript in prep.). Scaled production of a high-value chemical from microalgae under natural environmental conditions.

Tran, N. A. T., Tamburic, B., Evenhuis, C. R., & Seymour, J. R. (2020). Bacteria-mediated aggregation of the marine phytoplankton *Thalassiosira weissflogii* and *Nannochloropsis oceanica*. *Journal of Applied Phycology*, 32(6), 3735-3748.

Holzer, A., Newsad, S., **Tran, N. A. T.**, Harrison, E., Smith A.G (2020) Microalgae, bacteria and vitamins: three key players in aquatic microbial communities. *Microbiology Today*, 47(1): 51-53.

SELECT AWARDS & ACHIEVEMENTS

2025 MIT Leadership and Professional Strategies and Skills Course

2023 MIT J-WAFS seed grant (\$150,000)

2022 Travel Grant, Wolfson College Cambridge (£1200)

2021 Research Grant, Cambridge Philosophical Society (£2250)

2016 Chancellor's Research Scholarship, University of Technology Sydney (\$10,000)

2016 UTS University Medal (Awarded to the undergraduate student with the highest GPA in each graduating cohort and the highest level of achievement in an Honours course)